

HARNESS.md – CCAT Leash

Purpose & System Overview

This harness governs a CCAT-style question generation system. The **worker agent** creates and revises one practice question at a time (comprising a prompt, answer choices, one correct answer, and an explanation). The **harness** completely owns the surrounding governance: it determines final acceptance, runs checkpoints, executes tools, handles persistence, and monitors safety/operational metrics. The worker agent cannot decide when a question is acceptable.

User Request → Harness → Generation Request → Worker Agent → Candidate Question → Harness Checkpoints → Accepted/Revised/Rejected/Escalated

Core Loop & Checkpoints

The harness runs a continuous loop until a user's requestedQuestionCount is successfully reached. If a candidate fails a checkpoint, the harness returns structured CheckpointFeedback and permits a **maximum of 3 revisions** before emitting a MAX_REVISIONS_EXCEEDED alarm and forcing a new regeneration.

Checkpoint Validation Matrix

Checkpoint Name	Pass Criteria	Failure Behavior
Schema Check	Required fields are present, valid JSON, and structured	Reject and regenerate using schema feedback
Answerability Check	Question prompt has sufficient information to solve	Ask agent to revise stimulus or prompt wording
Single Answer Check	Exactly one answer choice is valid and correct	Ask agent to revise choices or prompt
Explanation Check	Reasoning fully supports the declared correct answer	Ask agent to correct/expand explanation logic
Math Check	Calculations strictly match deterministic tool output	Revise math calculation, choices, or regenerate
Similarity Check	Question does not closely paraphrase seed or prior examples	Reject completely and regenerate fresh candidate
Category Check	Layout maps accurately to the	Revise taxonomy targeting or

	requested taxonomy	relabel category
Difficulty Check	Complexity aligns reasonably with requested target	Revise or relabel difficulty
Content Safety Check	Content is appropriate for a general study tool	Reject or revise offensive/unsafe entries
Export Readiness	Final record can be cleanly packaged to output schemas	Fix structure to ensure clean output serialization

Technical Interfaces & Tooling

The harness treats the worker agent as completely replaceable via a swappable interface supporting two core operations:

1. generateQuestion(input: GenerationRequest): CandidateQuestion
2. reviseQuestion(candidate: CandidateQuestion, feedback: CheckpointFeedback[]): CandidateQuestion

All data flowing through the loop is rigorously structured as programmatic objects. The worker never executes tools directly; it must request actions through the harness, which intercepts, executes, and records them into a ToolCallTrace object. Approved tools include an exact arithmetic calculator, deterministic math checkers, similarity detectors, and formatting validators.

Observability, Alarms, & Escalation

The system maintains strict observability across two data layers: **Quality Traces** (logging every candidate version, tool call trace, guardrail failure, and rejection reason) and **Operational Metrics** (tracking token usage, cost limits, latency, and checkpoint failure frequencies).

The harness will instantly halt automated processing and bundle a HumanReviewPackage for manual escalation when:

- Tool outputs conflict directly with agent explanations.
- A single checkpoint type repeatedly fails or a similarity score remains borderline.
- Configured financial cost thresholds or token limits are approaching.
- The system-wide acceptance rate falls below a designated threshold.

Implementation Priorities

- **Phase 1 — Basic Harness:** Enforce the minimum working loop. Run basic validation (schema, single answer, explanation), accept or reject candidates, and persist successful questions directly to a local JSON layout.
- **Phase 2 — Tool-Based Validation:** Implement the integration of external deterministic tools (exact arithmetic calculator, math verifier, similarity checking engines).
- **Phase 3 — Revision Loop:** Deploy the structured feedback array mechanism, enabling the swappable worker to process specific checkpoint failures within the revision ceiling.
- **Phase 4 — Observability:** Build the logging and monitoring architecture to compute real-time operational metrics, token consumption, and cost tracking.
- **Phase 5 — Alarms & Escalation:** Add final production safety controls, structured alarm handling, and automated human-in-the-loop packaging templates.

Final Design Principle: The worker agent creates. The harness governs. The harness must be deterministic wherever possible, explicit in every checkpoint, conservative in acceptance, and complete in traceability. No question is accepted simply because it looks good.